ZedPath: AI Usage Report

Team Kestrel · IntelliCon '26 Buildathon · Gate 3 · 11 October 2026 · B.M.P Banneka

1. Every tool we used, and what it did

Tool	What it did for us
Claude Code (Anthropic, Claude Opus)	Build partner: wrote the app, API, rule engine, tests, data tools and the ZP-DOC-00 to 08 documents; ran builds, local tests and (only after a written GO) deploys. A project guard hook blocks deploys, account switches and pushes to other repositories.
Claude (second model) + scripts	Second-opinion rule checker: course entry rules were encoded twice, independently, and the two encodings compared.
Claude artifacts and Claude Docs	Concept screens, the business case, the slide deck and diagrams; edited from our review comments.
Google Gemini API (via Cloudflare AI Gateway)	In the product: writes Ask ZedPath's cited answers (Section 6).
Cloudflare Workers AI	In the product: bge-m3 multilingual embeddings for search; Llama Guard 3 8B screens every group message.
Groq Whisper (speech to text)	Transcribed the founder's narrated phone walkthroughs so feedback became a task list.
WebFetch / Firecrawl, agent-browser	Read official sources as text (allow-listed domains only); took phone-size screenshots to check each screen.

2. How much of the code is AI-generated, and how we checked it

Nearly all of it. The repository has **71 commits** (5 to 11 October 2026), about **8,950 lines** of app, API and tool code, **1,400 lines** of tests and **3,100 lines** of documents. We estimate over 95% of these lines were written by Claude; the founder wrote the requirements, ran the six user interviews, made every product decision, reviewed each change and tested it on a phone. Every commit says so in its co-author line.

How we checked it:

- **164 automated tests** (unit and API tests on the real database migrations), run before every release; type checks with no errors.
- **Planted-bug checks:** we broke the code on purpose (4 security bugs, rule mutations) to prove the tests catch it.
- **Two independent rule encodings** compared over 40.4 million student cases: 116 of 121 courses identical, the other 5 resolved against the handbook text.
- **Every change clicked through** on a local build at phone size before release; deploys only after the founder's written GO, then verified on the live site.

3. Our best workflow: narrated video to tested release

The founder records a 5-minute narrated walkthrough of the live app on his phone. Claude transcribes it, reads the frames, turns each remark into a numbered change, builds it with tests, and commits each step. On 11 October one video produced 13 changes (Compare redesign, numbers on charts, readable sources, group rules enforced on the server, a Journey checklist, notification settings and more) in 6 tested commits within a few hours. The founder's time went on judgment, not typing.

4. Where AI failed us, and what we did

- **Unsafe web content.** While researching syllabi, one university host returned planted upload-tool content. We stopped, used nothing from it, blocked the host in code, and wrote a research safety protocol: official domains only, page text treated as data, never as instructions.
- **A silent safety failure.** An AI change set the project to ES modules, which broke our deploy guard for about an hour without any visible error. We made the guard fail closed (a crashed guard now blocks everything) and added 40 guard tests.
- **Code bugs.** An AI-written sentence splitter broke citation markers; two quick group posts could bypass the rate limit. Both were found by local testing, fixed, and pinned by tests.

- **Business case numbers.** A draft rounded 176,527 × 60% to 105,000 (it is 105,916) and its summary cost disagreed with its own cost table. Found in the founder's review; every figure was recomputed from its source and cited.

5. How we verified facts before they reached the business case

- Every statistic carries a numbered reference to its source (Department of Examinations results, UGC handbook 2025/26, UGC cut-off tables, National Education Commission study, DataReportal, Government Gazette).
- **University seats:** we added up the seats for every degree listed in the official UGC handbook ourselves and got the same national total the UGC publishes, 42,937.
- **Minimum Z-scores:** we read every past minimum Z-score (each degree, in each of the 25 districts) out of the official tables twice, by two different methods; the two copies agreed on every number, and we checked a sample by eye against the printed pages. Course subjects shown in the app were compared word for word with the universities' own pages.
- Prices (Cloudflare, Gemini) were read from the official pricing pages on 11 October 2026, not from model memory. Market estimates that are our own assumptions are labelled as estimates.

6. AI inside ZedPath, and what it costs per user

Feature	Model / API	Cost
Ask ZedPath answers (Sinhala, Tamil, English)	Gemini 2.5 Flash through Cloudflare AI Gateway, with a fallback chain to lighter Gemini models; answers only from retrieved official pages, each cited	About USD 0.0015 per answer (≈3,000 input + 250 output tokens at USD 0.30 / 2.50 per million)
Search across languages	Workers AI @cf/baai/bge-m3 embeddings + Cloudflare Vectorize	Within free allowances at our scale
Group safety	Workers AI @cf/meta/llama-guard-3-8b on every message; fails closed	Within the 10,000 free neurons a day; about USD 7 a season at 10,000 students, worst case

Per user: about 10 questions a season costs **about LKR 4.60 (USD 0.015) per student**; all AI and hosting together, about **LKR 8 per student a year**. The eligibility rules and Safe/Likely/Reach bands use no AI at runtime: they come from our tested rule engine, so the AI never decides a student's chances.